

Description

The potato (*Solanum tuberosum*) is a widely cultivated root vegetable in the Solanaceae family. They are native to the Andes Mountains in South America and have been grown for thousands of years. Potatoes are a staple food in many countries around the world and are used in a wide variety of dishes. They are high in carbohydrates, fiber, and vitamins C and B6.

Potatoes are typically planted in the spring and harvested in the late summer or early fall. They grow best in well-drained soils with a pH between 4.5 and 7.5. Potatoes require a lot of water, especially during the flowering and fruiting stages, and they are sensitive to frost.

The crop can be propagated from seed potatoes, which are small tubers that are planted in the soil. They can also be propagated from potato seed, which is a less common method. Once the seed potato or seed is planted, the plant will grow leaves and flowers, and then it will produce tubers underground.

The potato is a herbaceous perennial plant grown for its edible tubers. The plant has a branched stem and leaves consisting of leaflets that are both unequal in size and shape and can be oval to oblong in shape. The leaves can reach up to 10–30 cm in length. Tubers grow about 25 cm underground in the soil and can be yellow, red, or purple depending on the variety. The potato plant produces white or blue flowers and yellow-green berries. The plants can go up to 1m (3.3ft) in height.

Potatoes are grown in many countries across Africa, but the main potato-producing countries are Egypt, South Africa, Ethiopia, and Kenya. In these countries, potatoes are grown both for local consumption and for export.

Potatoes are a relatively new crop in Africa, having been introduced to the continent in the late 19th century. However, since then, potato production has increased significantly, with farmers in Africa recognizing the potential of the crop as a food security and income generation opportunity.

Potatoes grow well in the cool highland regions of Africa, where the temperature and rainfall are suitable for the crop. However, potato cultivation in Africa is often challenged by a lack of access to quality seed potatoes, a lack of irrigation systems, and the prevalence of pests and diseases.

FAOSTAT reports that production of potatoes in Africa has more than doubled, with eastern Africa producing the biggest yield. The crop is primarily produced by small-holder farmers, with Algeria leading in production with 4,928,028 tons per year, overtaking Egypt, which produces 4,800,000 tons per year.

The following is a summary of the physical attributes of potato crops:

Leaves: The leaves of the potato plant are simple and alternate. They are generally dark green and glossy in color and are smooth or slightly hairy.

They are lobed or ovate in shape and can grow up to 15 cm long.

Stems: The stems of the potato plant are green, smooth, and slightly hairy.

They grow to a height of 30–90 cm and are usually unbranched.

Flowers: The potato plant produces small, white or purple flowers that grow in clusters. These flowers are typically about 2 cm in diameter and have five petals.

Tubers: The tubers are the underground storage organs of the potato plant and are the part of the plant that is typically consumed as a vegetable.

They are round or oblong in shape and can vary in size and color depending on the variety. They can be white, yellow, red, or purple.

Root: The potato plant has a deep taproot and fibrous root system that allow the plant to absorb water and nutrients from the soil.

Fruits: Potato plants produce small, green, and poisonous berries that are typically not consumed by humans.

Crop Details

Scientific Name: *Solanum tuberosum*

Common Name: Viazzi mbatata (Swahili), Waru (Kikuyu), Papa (Spanish), Kartoffel (German), Patata (Italian).

Uses & Benefits

Potato tubers are a staple food source in temperate regions and are eaten after cooking. They may be cut or sliced and made into potato chips or fries. Potatoes can also be processed into starch, alcohol or flour.

Potatoes are a good source of carbohydrates, potassium, and vitamin C; they are also a source of fiber, vitamin B6, and folate. They are low in fat, calories, and sodium.

The values shown in the following table are approximate and may vary depending on the variety, growing conditions, and preparation method of the potatoes. Additionally, cooking and processing methods such as frying or mashing can change the nutritional value of potatoes and increase their calorie and fat content.

Varieties

Potatoes are grown in many countries across Africa, and different varieties are grown depending on the region and the local market. For example, in Egypt, the most common variety grown is Desiree, which is a red-skinned variety with a waxy texture that is good for boiling and mashing. In South Africa, farmers grow varieties such as "Bintje," "Sifra," and "Agria," which are known for their high yields and good storage capabilities. In Ethiopia, farmers grow varieties such as "Sarpo Mira," which is known for its resistance to pests and diseases, and "Shepody," which is known for its good taste and texture. In Kenya, farmers grow varieties such as "Kenya Mpya," "Shangi", and "Kenya Raha," which are known for their high yields and good taste.

It's worth noting that Africa has a rich diversity of local varieties of potatoes known as "landraces." These varieties have been developed and adapted to the local conditions and have the potential to provide food security and income for the farmers. However, the use of improved varieties and the adoption of good agricultural practices are necessary to achieve the full potential of potato production in Africa.

Propagation

Basic Requirements

Potatoes are a cool-season crop that grows best in certain climatic, soil, and water conditions.

Potatoes prefer a cool climate with moderate temperatures and a long growing season. They grow best at temperatures between 10–15 °C (50–59 °F) during the day and 5–10 °C (41–50 °F) during the night. Potatoes are sensitive to frost,

and temperatures below 0°C (32°F) can damage the tubers. Potatoes also require a moderate-to-high amount of rainfall or irrigation.

They grow best in well-drained soils that are rich in organic matter and have a pH between 4.5 and 7.5. They prefer soils that are high in nitrogen, phosphorus, and potassium and also require adequate levels of calcium, magnesium, and sulfur. Potatoes also grow well in sandy loam or clay soils.

Potatoes require a lot of water, especially during the flowering and fruiting stages. They need about 1-2 inches of water per week, and more in hot and dry weather. They prefer consistent moisture throughout the growing season but also require good drainage to prevent waterlogging and root rot. Potatoes are sensitive to drought and water stress, which can cause reduced yields and lower-quality tubers.

It's worth noting that while these are the ideal conditions for growing potatoes, the crop can be grown in a wide range of conditions, and some varieties are more tolerant to different conditions than others. Additionally, agronomic practices such as fertilization, irrigation, and crop rotation can help to improve the growth and yields of potatoes in challenging environments.

Growing from Seed

The planting procedure for potatoes generally includes the following steps:

Preparing the soil: The soil should be well-drained and have a pH between 4.5-7.5. Prior to planting, the soil should be worked to a fine tilth, and any necessary amendments such as fertilizer or lime should be added.

Choosing seed potatoes: Seed potatoes should be disease-free and certified. They should be cut into pieces with at least one "eye" per piece, and should be left to dry for a few days before planting to allow the cuts to heal over.

Determining the planting date: The best time to plant potatoes will depend on your climate and the variety you are growing. Potatoes are typically planted in the spring, after the last frost date, and when the soil has warmed up to around 10-15°C (50-59°F).

Planting the seed potatoes: Seed potatoes should be planted at a depth of around 10-15 cm, with the "eyes" facing upward. They should be spaced around 30-40 cm apart, in rows that are 75-90 cm apart.

Watering and mulching: After planting, the soil should be watered well to settle the soil around the seed potatoes and to encourage germination. A layer of mulch, such as straw or grass clippings, can be applied to help retain moisture and suppress weeds.

General Care and Maintenance

As the plants grow, they will require regular watering, fertilization, and weeding. The plants should be hilled up (soil should be added to the base of the plant) as they grow to prevent the tubers from being exposed to sunlight, which can cause them to turn green and become inedible.

Harvesting

Harvesting potatoes is the process of collecting the tubers, which are the edible part of the potato plant, from the soil. The timing of the harvest is crucial and will depend on the variety of potato being grown. Generally, potatoes are ready to be harvested when the foliage has died back and the skin of the tubers has hardened.

Before harvesting, farmers need to check the tubers for maturity. This can be done by gently digging up a few plants and checking the size and skin of the tubers. The tubers should be fully formed, and the skin should be tough enough that it can't be easily scraped off with your fingernail.

When it comes to the actual harvesting process, farmers should be careful not to damage the tubers. They should be carefully dug up using a spade or potato fork, taking care not to damage the tubers. It's better to harvest when the soil is dry and not wet. After harvesting, the tubers should be placed in a warm, dark, and well-ventilated place for a week or two to "cure" (harden the skin) before being stored.

After curing, the tubers should be cleaned of any dirt and debris, and any damaged or diseased tubers should be removed and discarded. Potatoes should be stored in a cool, dark, and well-ventilated place with a temperature between 4 and 8 °C (39 and 46 °F) and a relative humidity of 90 to 95 %.

It's important to harvest potatoes at the right time to ensure the best quality and storage potential. Overripe potatoes have a higher chance of spoilage and can't be stored as long as mature ones. Additionally, it's also important to properly

cure, clean, and store the tubers to ensure they maintain their freshness and quality for as long as possible.

Common Pests and Diseases

Diseases

Category : Bacterial

Bacterial ring rot *Clavibacter michiganensis*

Symptoms

Wilting stems and leaves; dying leaves; lower leaves wilting first; ring of creamy yellow to brown rot visible when tuber is cut crossways

Cause

Bacterium

Comments

Bacterium is tuber-borne; bacteria can enter tuber through cutting wounds; disease favored by wet, warm soils; bacteria overwinter in potato debris

Management

Plant only certified seed potatoes - certified seed potatoes are grown in seed beds with zero tolerance of ring rot; remove all crop debris from soil after harvest; sanitize tools and equipment regularly

Blackleg (Soft rot) *Erwinia carotovora*

Symptoms

Small, water-soaked lesions on base of stems originating from seed piece; lesions may enlarge to form a large extended lesion stretching from base of stem to canopy; tissue becomes soft and water-soaked and can be light brown to inky black in color; wilted, curled leaves which have a soft and slimy texture when wet

Cause

Bacterium

Comments

Bacteria are carried on tubers and in wounds and can be spread to healthy tubers during handling and cutting of seed pieces; disease emergence favors high soil temperatures

Management

Plant seed pieces which are the product of tissue culture; sanitize tools and equipment when cutting seed pieces to prevent bacterial contamination; avoid damaging tubers during harvest; reduce periods of leaf wetness by allowing enough time for leaves to dry throughout the day after watering

Common scab *Streptomyces* spp.

Symptoms

Raised brown lesions on tubers with corky texture; deep, pitted brown or black lesions on tuber with straw-colored translucent tissue underneath

Cause

Bacterium

Comments

Disease is most severe during warm and dry conditions

Management

Common scab can be very difficult to manage and prevention of the disease relies on combining several different methods. These include: avoiding planting infected tubers, using a 3-4 rotation away from potato; planting less susceptible potato varieties (none are immune); maintaining a high soil moisture content for 4-6 weeks after stolon tips begin to swell at the onset of tuber development; amending soil to lower pH and treating seed with appropriate fungicides when available.

Category : Fungal

Black dot *Colletotrichum coccodes*

Symptoms

Small black dots (fungal fruiting bodies) on tubers, stolons and stems; roots may rot below ground; leaves may turn yellow and wilt; infection may cause defoliation

Cause

Fungus

Comments

Disease emergence favors poorly draining soil; poor aeration of soil and high temperatures; disease symptoms are most severe in coarse soils that are low in nitrogen

Management

Stressed plants are more susceptible to black dot; rotate crops away from potato; do not plant infested tubers or seed pieces; fertilize and water plants adequately; protective fungicides can be applied where available.

Black scurf & Rhizoctonia canker *Rhizoctonia solani*

Symptoms

Flat, irregularly shaped black or dark brown fungal fruiting bodies on tuber surface; tubers may be misshapen; red-brown to black sunken lesions on sprouts; lesions may girdle the main stem causing leaves to curl and turn yellow

Cause

Fungus

Comments

Fungus can be spread by infested soil or planting infected seed pieces and tubers; disease emergence favors cool, moist soil

Management

No potato varieties are completely resistant to the disease; control relies on reducing the level of inoculum in both the soil and in tubers; methods include applying fungicide to seed pieces or soil; avoiding planting seeds too deeply in cold soils and rotating crop away from potato to reduce levels of soil inoculum.

Gray mold *Botrytis cinerea*

Symptoms

Flowers covered in gray, fuzzy mold; wedge shaped tan lesions on leaves; a slimy brown rot may be present on stems, originating from the petiole; infected tubers have wrinkly skin and tissue underneath is soft and wet; tubers often develop a gray fuzzy growth

Cause

Fungus

Comments

Disease emergence favors excessive humidity, cool temperatures and shade

Management

Cultural control is very important for managing the disease, provide plants with adequate fertilizer and water; application of appropriate protective fungicides where available can provide adequate control of disease but will not treat an established infection

Pink rot *Phytophthora erythroseptica*

Symptoms

Stunted plant growth; wilting leaves; dying leaves; marked tuber decay; dark brown eyes on tuber; cut tuber turns pink after 20-30 min air exposure, then turns brown and finally black

Cause

Oomycete

Comments

Disease emergence favors high soil water saturation late in the season

Management

Plant potato in well-draining soils with no history of pink rot; avoid overwatering plants; avoid wounding during harvest

Potato Early Blight *Alternaria solani*

Symptoms

Dark lesions with yellow border which may form concentric rings of raised and sunken tissue on the leaves and stems; lesions initially circular but become angular; leaves become necrotic but remain attached to plant; dark, dry lesions on tubers with leathery or corky texture and watery yellow0green margins

Cause

Fungus

Comments

Disease emergence favors cycles of wet and dry conditions with periods of high humidity and leaf wetness

Management

Application of appropriate protective fungicide can reduce severity of foliar symptoms; reduce stress to plants by fertilizing and watering adequately; plant late varieties which are less susceptible to disease; store tubers in cool environment

Powdery scab *Spongospora subterranea***Symptoms**

White to brown galls on the roots and stolon; raised pustules on tuber surrounded by potato skin; shallow depressions on tuber filled with brown spores

Cause

Fungus

Comments

Symptoms all occur below ground

Management

Do not plant tubers showing symptoms of disease; avoid planting potato in poorly draining soils; if disease occurs rotate crop away from potato for a period of 3-10 years and avoid planting tomato

Verticillium wilt *Verticillium dahliae*

Verticillium albo-atrum

Symptoms

Early death of plants; leaflets dying on only one side of the petiole or branching stem; cut through the stem reveals a discoloration of the tissue; discoloration of tubers at stem-end

Cause

Fungi

Comments

Disease emergence favors high temperature and moisture early in season followed by drought; disease can be spread to uninfected fields by wind or movement of infested soil particles

Management

Planting resistant varieties of potato is the most common method of controlling the disease; cultural practices such as using furrow irrigation in place of sprinkler irrigation and avoiding overwatering plant can also reduce the severity of the disease should it occur

Category : Other

Leak *Pythium* spp.

Symptoms

Light tan, water soaked area around wound on tuber; internal rotting of tuber which results in internal tissue becomes spongy and possibly developing cavities; dark, watery fluid exudes from the tuber when squeezed

Cause

Fungi

Comments

Disease only affects tubers and fungus can only enter through wounds; all common potato cultivars are susceptible to leak; disease emergence is favored by relatively high temperatures

Management

Rotating crops away from potato and destroying any infected tubers helps to control the disease; infection can be reduced by application of appropriate foliar fungicides; delaying harvest to allow the skin of the tuber to mature reduced the risk of injury to the tuber

Potato Viral Disease

Symptoms**Cause****Comments**

Category : Oomycete

Potato Late Blight *Phytophthora infestans*

Symptoms

Irregularly shaped spreading brown lesions on leaves with distinctive white fluffy sporulation at lesion margins on the underside of the leaf in wet conditions. In dry condition the lesions dry up and go dark brown with collapsed tissue; water-soaked dark green to brown lesions on stems also with characteristic white sporulation; later in infection leaves and petioles completely rotted; severely affected plants may have an slightly sweet distinctive odor; red-brown firm lesions on tubers extending several centimeters into tissue; lesions may be slightly sunken in appearance and often lead to secondary bacterial rots.

Cause

Oomycete

Comments

The pathogen can survive for several months to years in the soil; emergence of disease favored by moist, cool conditions; major cause of disease spread is infected tubers

Management

Control depends on a multifaceted approach with importance of certain practices changing based on geographic location: destroy infected tubers; destroy any volunteer plants; application of appropriate fungicide to potato hills at emergence; time watering to reduce periods of leaf wetness e.g. water early to allow plant to dry off during the day; plant resistant varieties; apply appropriate protective fungicide if disease is forecast in area

Category : Viral

Potato leaf roll Potato leafroll virus (PLRV)

Symptoms

Young leaves rolled and yellow or pink; lower leaves have leathery texture and roll upward; necrotic netting in vascular tissue of tuber may be present; plant exhibits an upright growth habit and growth may be stunted

Cause

Virus

Comments

Transmitted by several species of aphid; infected seed tubers and volunteer potato plants provide a source of inoculum for the virus

Management

Grow plants produced clonally from PLRV-free stock; harvest potato crop early in temperate regions to avoid aphid migrations late in season; remove and destroy infected plants and tubers; application of appropriate insecticides where available may help reduce spread

Potato virus A Potato virus A (PVA)

Symptoms

Mild mosaic pattern or mottling on leaves; severely infected plants may have alternating patches of yellow and dark green tissue; leaves may have a shiny appearance; stems bending outwards slightly

Cause

Virus

Comments

Virus is transmitted by several species of aphid and can be transmitted to the next potato generation by planting infected tubers; tubers show no visible symptoms

Management

Plant material which is free of PVA; plant tolerant potato varieties in areas where virus is common; application of appropriate systemic insecticides helps to control aphid populations`

Potato virus X Potato virus X (PVX)

Symptoms

Mild mosaic pattern on leaves; severely infected plants may be dwarved with smaller leaves; necrosis of plant tops and tubers may occur

Cause

Virus

Comments

PVX can be transmitted by infected leaves coming into contact with healthy ones

Management

Planting seed free of PVX is the most important method of controlling the virus

Potato virus Y Potato virus Y (PVY)

Symptoms

Symptoms vary widely from mild mosaic of leaves to leaf necrosis and plant death depending on the variety of potato and the strain of the virus: leaves may turn yellow and drop from plant; symptoms may be present on only one shoot of the plant; plants with severe leaf necrosis may produce tubers with light brown rings on the skin

Cause

Virus

Comments

Virus is transmitted by more than 25 different species of aphid; virus can be transmitted over long distances by aphids; can be transmitted mechanically by contact with infected leaves or tubers

Management

Plant only certified seed potatoes; application of appropriate systemic insecticides, where available, may control spread of virus within a single plantation but will not prevent spread caused by winged aphids; remove and destroy symptomatic plants; plant resistant varieties

Pests

Category : Insects

Aphids (Peach aphid, Potato aphid) *Myzus persicae*

Macrosiphon euphorbiae

Symptoms

Small soft bodied insects on underside of leaves and/or stems of plant; usually green or yellow in color, but may be pink, brown, red or black depending on

species and host plant; if aphid infestation is heavy it may cause leaves to yellow and/or distorted, necrotic spots on leaves and/or stunted shoots; aphids secrete a sticky, sugary substance called honeydew which encourages the growth of sooty mold on the plants

Cause

Insects

Comments

Aphids are most damaging to potato through the transmission of viruses such as Potato leafroll virus; distinguishing aphid features include the presence of cornicles (tubular structures) which project backwards from the body of the aphid; will generally not move very quickly when disturbed

Management

If aphid population is limited to just a few leaves or shoots then the infestation can be pruned out to provide control; check transplants for aphids before planting; use tolerant varieties if available; reflective mulches such as silver colored plastic can deter aphids from feeding on plants; sturdy plants can be sprayed with a strong jet of water to knock aphids from leaves; insecticides are generally only required to treat aphids if the infestation is very high - plants generally tolerate low and medium level infestation; insecticidal soaps or oils such as neem or canola oil are usually the best method of control; always check the labels of the products for specific usage guidelines prior to use

Colorado potato beetle *Leptinotarsa decemlineata*

Symptoms

Feeding damage to foliage; if infestation is severe or if left untreated plants can be completely defoliated; adult insect is a black and yellow striped beetle; larvae are bright red with black heads when they first hatch and change color to pink; larvae have two rows of black spots

Cause

Insect

Comments

Adult beetles emerge in spring; female beetles lay eggs in batches of up to two dozen; eggs are orange-yellow and are laid on undersides of leaves; a female can lay 500 or more eggs over a four to five week period

Management

Control of Colorado potato beetle can be challenging as they have developed high levels of insecticide resistance; in the home garden planting early maturing varieties of potato allows the plants to escape from most damage; adults and larvae should be hand picked from plants and destroyed in soapy water; applications of *Bacillus thuringiensis* can be effective at controlling larvae but should be applied frequently; some insecticides, including spinosad, are still effective against adult beetles

Cutworms *Agrotis* spp.

Peridroma saucia

Nephelodes minians

and others

Symptoms

Stems of young transplants or seedlings may be severed at soil line; if infection occurs later, irregular holes are eaten into the surface of fruits; larvae causing the damage are usually active at night and hide during the day in the soil at the base of the plants or in plant debris of toppled plant; larvae are 2.5–5.0 cm (1–2 in) in length; larvae may exhibit a variety of patterns and coloration but will usually curl up into a C-shape when disturbed

Cause

Insects

Comments

Cutworms have a wide host range and attack vegetables including asparagus, bean, cabbage and other crucifers, carrot, celery, corn, lettuce, pea, pepper, potato and tomato

Management

Remove all plant residue from soil after harvest or at least two weeks before planting, this is especially important if the previous crop was another host such as alfalfa, beans or a leguminous cover crop; plastic or foil collars fitted around plant stems to cover the bottom 3 inches above the soil line and extending a couple of inches into the soil can prevent larvae severing plants; hand-pick larvae after dark; spread diatomaceous earth around the base of the plants (this creates

a sharp barrier that will cut the insects if they try and crawl over it); apply appropriate insecticides to infested areas of garden or field if not growing organically

Flea beetles *Epitrix* spp.

Symptoms

Small holes or pits in leaves that give the foliage a characteristic “shothole” appearance; young plants and seedlings are particularly susceptible; plant growth may be reduced; if damage is severe the plant may be killed; the pest responsible for the damage is a small (1.5–3.0 mm) dark colored beetle which jumps when disturbed; the beetles are often shiny in appearance

Cause

Insects

Comments

Younger plants are more susceptible to flea beetle damage than older ones; older plants can tolerate infestation; flea beetles may overwinter on nearby weed species, in plant debris or in the soil; insects may go through a second or third generation in one year

Management

In areas where flea beetles are a problem, floating row covers may have to be used prior to the emergence of the beetles to provide a physical barrier to protect young plants; plant seeds early to allow establishment before the beetles become a problem - mature plants are less susceptible to damage; trap crops may provide a measure of control - cruciferous plants are best; application of a thick layer of mulch may help prevent beetles reaching surface; application on diamotecoous earth or oils such as neem oil are effective control methods for organic growers; application of insecticides containing carbaryl, spinosad, bifenthrin and permethrin can provide adequate control of beetles for up to a week but will need reapplied

Wireworms *Aeolus* spp.

Anchastus spp.

Melanotus spp.

Limonius spp

Symptoms

Death of seedlings; reduced stand; girdled stems and white heads; wireworm larvae can be found in soil when dug round the stem; larvae are yellow-brown, thin worms with shiny skin

Cause

Insect

Comments

Larval stage can last between 1 and 5 years depending on species

Management

Chemical control impossible in a standing crop, must be applied at preplanting or as a seed treatment; if wireworms are known to be present in soil fallow field during summer and till frequently to reduce numbers; rotate to non-host crop where possible; avoid planting susceptible crops after a wireworm infestation on cereals without either fallowing or applying appropriate pesticide